

1. Question:

Write a Python program that calculates the sum of all even numbers within a user-defined range (inclusive). The program should first ask the user for a starting number and an ending number. Then, it should iterate through all numbers in that range and add only the even numbers to a running total. Finally, it should print the total sum of even numbers found.

Output Example:

```
Enter the starting number: 1
Enter the ending number: 10
The sum of even numbers between 1 and 10 is: 30
```

2. Question:

Write a Python program that asks the user to enter a single word. The program should then count how many vowels (a, e, i, o, u - case-insensitive) are present in that word. Finally, it should print the total count of vowels found.

Output Example:

```
Enter a word: Hello
Number of vowels: 2
```

3. Question:

Write a Python program that asks the user to enter a positive integer. The program should then determine whether the entered number is a prime number or not. A prime number is a natural number greater than 1 that has no positive divisors other than 1 and itself. Finally, the program should print whether the number is prime or not.

Output Example 1:

```
Enter a positive integer: 7
7 is a prime number.
```

4. Question:

Write a Python program that asks the user for two positive integers: the **height** (number of rows) and the **width** (number of characters in each row) for a parallelogram. The program should then print a solid parallelogram pattern using asterisks (*), with leading spaces to create the slanted effect.

Output Example:

Enter the height of the parallelogram: 4

Enter the width of the parallelogram: 6

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*****
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*****
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*****
```

5. Question:

Write a Python program that asks the user for a single positive integer. This number will represent the "size" of the diamond (i.e., the number of stars in its widest row, and also the number of rows in its top half, including the widest row). The program should then print a solid diamond pattern using asterisks (*), with a single space between each star on the same line to create a distinct visual.

Output Example:

Enter the size of the diamond: 5

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